

Statement of Purpose (SOP)

Amazon ML Summer School 2026

I am Priya Kumari, third-year undergrad pursuing B.Tech in Computer Science and Engineering, from NIT Patna, expected to graduate in 2027. My interest in Machine Learning began during my coursework in Artificial Intelligence and Machine Learning, where I was introduced to the fundamental concepts that enable computers to learn from data and make intelligent decisions. Since then, I have been actively exploring ML through academic learning, projects, and self-study.

One of the most impactful experiences in my learning journey was developing a House Price Prediction project using Machine Learning techniques. Through this project, I worked on **data preprocessing, feature analysis, model training**, and performance evaluation to predict housing prices based on various factors. The project helped me understand how machine learning models transform raw data into actionable insights and strengthened my problem-solving and analytical skills. More importantly, it showed me the real-world impact of data-driven decision-making.

My academic curriculum has provided me with a foundation in Artificial Intelligence, Machine Learning, Data Structures and Algorithms, Database Management Systems, and other core computer science subjects. Beyond academics, I have actively sought opportunities to expand my technical skills. I participated in Smart India Hackathon 2024, where I experienced collaborative problem-solving and innovation in a competitive environment. I have also completed the Google Cloud Arcade Facilitator Program, which introduced me to cloud technologies and their role in building scalable intelligent applications. Currently, I am exploring projects that leverage technology to address real-world challenges. These experiences have strengthened my desire to pursue Machine Learning as a long-term career path. I am particularly interested in understanding how advanced ML techniques can be applied to solve large-scale problems in areas such as fraud detection, recommendation systems, predictive analytics, and intelligent automation.

Amazon ML Summer School presents a unique opportunity to learn from leading scientists and researchers who are working on cutting-edge machine learning applications. I am eager to deepen my understanding of core ML concepts, mathematical foundations, and industry best practices while gaining insights into how machine learning is applied at scale. The program's focus on both theoretical knowledge and practical applications aligns perfectly with my learning goals. My long-term aspiration is to build intelligent and trustworthy AI solutions that address societal and technological challenges, particularly in areas such as fraud detection,

cybersecurity, and intelligent automation. I view Amazon ML Summer School as an important step toward this goal, as it will provide me with the knowledge, mentorship, and exposure necessary to grow as a machine learning practitioner. I am eager to contribute my enthusiasm, dedication, and problem-solving mindset to the program while learning from peers and mentors who share a passion for machine learning. I believe this opportunity will significantly shape my journey toward becoming an AI/ML engineer capable of creating impactful solutions.

Thank you for considering my application.